CNN-Powered Brain Tumor Classification and Detection Using MRI based on Deep Learning Methods

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***Abstract*—It is very essential to use magnetic resonance imaging (MRI) to tell the differences between brain tumors, so that doctors could make a quick diagnosis and plan treatment. The study presents an automated model for the detection and classification of brain tumors using a Convolutional Neural Network (CNN). The proposed method prepares MRI images, identifies significant features, and categorizes tumors into four distinct groups. There are a lot of convolutional layers in the CNN architecture, and they use batch normalization and max pooling. Then it has dense layers with dropout regularization. We used a publicly available MRI dataset of brain tumors to train and test the model. The method is much better at classifying tumors than traditional machine learning models, as shown by experimental results.**

***Index Terms*—Brain Tumor, MRI, Classification, Convolutional Neural Network, Deep Learning**

I. Introduction

Brain Tumor (BT) Classification using MRI images has become an very important area of research because of its implementation in accurate treatment planning, early diagnosis and improvement in patient survival rate [1]. The abnormal growths of tissue in the brain is known as BT. These are classified into many categories like, meningioma, glioma and pituitary tumors. Traditional diagnosis techniques are very effective but they are dependent heavily on radiologist experts. To overcome these challenges and problems, researchers have moved to AI and deep learning (DL) based detection systems. Convolutional Neural Networks (CNNs) [2], are recognized for extracting features like hierarchical spatial features from image data, have gained great momentum just like a potent tool in the field of medical imaging (MI). The application of CNNs for BT classification has greatly improved diagnostic accuracy because it extracts appropriate features from MRI datasets [14] and reduce dependency on handcrafted feature engineering [3]. Various studies have proved the superior performance of CNNs while dealing with high-dimensional image data and learning non-linear structures, it also surpasses conventional machine learning (ML) algorithms like Support Vector Machines (SVMs), Random Forests (RFs), and K-Nearest Neighbors (kNN) [4] [5]. Data augmentation techniques such as contrast modification, rotation, flipping have also been proven very important in maximizing generalization and reducing overfitting to unseen data [6]. Inspired by these developments, our work formulates a good pipeline of DL based on CNN architectures to classify BTs into three groups from augmented and preprocessed MRI scans.

The structure of the paper is arranged as follows: Section 2 focuses on the works related to the paper. Section 3 dives deep into the proposed work. Section 4 gives a brief explanation on the results. Section 5 and 6 finally discusses and concludes the paper respectively.

II. Related Works

The application of DL in medical image analysis, particularly for BT classification, has been extensively explored over the past decade. Aggarwal et al. [7] proposed one of the early CNN-based models that demonstrated promising results in tumor segmentation using MRI modalities. Subsequent studies by Srinivasan et al. [8] further established the effectiveness of CNNs in both binary and multi-class classification tasks. Nahiduzzaman et al. [9] presented comparative analysis of CNNs' capabilities to excel classical ML models in identifying faint tumor patterns in MR images. The latest models have proposed hybrid structures like the ensemble-based structure by Bogacsovics et al. [10] and the Lightweight-CancerNet model by Raza et al. [11], which are aimed at delivering high performance with low computational overhead—thus being

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appropriate for real-time diagnosis. Enhanced networks such as Capsule Networks [12] and Transformer-based integrated architectures including TransBTS [13] have further extended boundaries by integrating spatial encoding with attention mechanisms for more detailed feature extraction. Together, these developments underscore the expanding influence of DL for BT diagnosis and form the basis upon which we construct our individual CNN-based classification model. In recent years, MRI image-based Classification of BTs has gained prominence due to its advantages of early diagnosis, precise treatment planning, and enhanced patient survival rates. BTs, which includes meningiomas (MEs), gliomas(GLs), and pituitary tumors (PTs), are abnormal growth of tissue. Although traditonal techniques depend greatly on the skills of radiologists and human-based MRI scan analysis, which is slow and may cause errors. Researchers have utilized DL and artificial intelligence (AI) to create automated, scalable, and accurate tumor detection systems to make it reliable.

III. Methodology

The proposed method consists of four key stages: data preprocessing, feature extraction, CNN-based classification, and model evaluation.

A. Data collection

In the study, a publicly accessible dataset from Kaggle, called the "Brain Tumor MRI Dataset" [14] was used, to enable reliable and accurate DL-based BT classification. Figshare, SARTAJ, and Br35H are three well-known sources that have been carefully combined to create the dataset, which is intended especially for multi-class BT classification tasks. It includes 7023 labeled MRI pictures of human brains that are divided into four groups: Pituitary Tumor (PT), Meningioma (ME), Glioma (GL), and No Tumor (NT). The Br35H dataset was the main source of the NT images, whereas the SARTAJ dataset's inaccurate or misclassified GL images were eliminated and swapped out for more precise samples from Figshare. The dataset is divided into training and testing sets. The training set included 1,321 images of GL, 1,339 images of ME, 1,457 images of PT and 1,595 images of NT. The testing set included 300 images of GL, 306 images of ME, 300 images of PT and 405 images of NT. Image preprocessing was very important because of the dataset's heterogeneity. Many of the original images had different sizes and they can also have unnecessary margins. To maximize model accuracy and guarantee consistency , each image underwent preprocessing which included margin removal, resizing and normalization. For both detection and classification tasks, as well as tumor localization using segmentation techniques, the dataset facilitates the use of CNN-based models. With more than 129,000 downloads and more than 22.3K views in the last 30 days, this dataset has emerged as a standard in recent research, demonstrating its popularity and relevance in the fields of ML and MI. The data's rich diversity and practical applicability make it best for developing DL models to classify BTs with high accuracy, leading to more effective and scalable medical diagnostics.

B. Data Preprocessing

All the MRI images were standardized to the same size by resizing MRI images to (128,128,3). To make the dataset more variable, augmentation techniques like rotation, contrast adjustment and flipping are also used. For multi-class classification, the labels are encoded into four classes using one-hot encoding. The dataset is divided into three parts: training, validation, and testing. The parts are 80%, 10%, and 10% of the whole. Data augmentation is very important to make it easier to work with new data and stop overfitting. Histogram equalization is also used to make the contrast of image better and make it easier to find features.

C. Feature Extraction

A crucial step in the CNN-based BT classification model is feature extraction. It converts unprocessed input images into useful representations that can be applied to precise classification. To extract hierarchical spatial features, each image $\mathbf{I} \in \mathbb{R}^{128\times128\times3}$ is subjected to a number of convolutional layers, activation functions, batch normalization, and pooling layers. Let $\mathbf{I}$ represent the input image and $\mathbf{K}$ represent a learnable kernel (filter) of size $k \times k$. The following is an expression for the convolution operation in the $l-th$ layer: $\mathbf{Z}^{(l)} = \mathbf{I}^{(l-1)} * \mathbf{K}^{(l)} + \mathbf{b}^{(l)}$ where $\mathbf{b}^{(l)}$ is the bias term and $*$ indicates the convolution operation.

Next, an activation function is used, like ReLU (Rectified Linear Unit):

$$\mathbf{A}^{(l)} = \text{ReLU}(\mathbf{Z}^{(l)}) = \max(0, \mathbf{Z}^{(l)})$$

Max pooling is used as follows to minimize spatial dimensions and manage overfitting:

$$\mathbf{P}_{i,j}^{(l)} = \max_{(m,n)\in\Omega} \mathbf{A}_{i+m,j+n}^{(l)}$$

where the pooling region is Ω (e.g., 2×2).

The arrangement of these layers is as follows:

Conv2D item $\rightarrow$ ReLU $\rightarrow$ MaxPooling $\rightarrow$ BatchNormalization $\rightarrow$ Repeated with increasing filter sizes: 32, 64, and 128 After being flattened into a feature vector $\mathbf{f} \in \mathbb{R}^d$, the finished feature maps are sent to fully connected layers for classification. The model can learn intricate patterns like tumor shapes, textures, and locations thanks to this hierarchical feature extraction, which is essential for differentiating between GL, ME, PT and NT categories.

D. Proposed DL Model

In this work,a CNN architecture was built, basically designed for the multi-class classification of BTs from magnetic resonance imaging (MRI). To implement the model we used TensorFlow Keras framework which has a layer-by-layer sequential structure. Resized and preprocessed MRI images with dimensions of $(128 \times 128 \times 3)$, which stand for the height, width, and color channels (RGB) respectively, are accepted by the input layer.

Three convolutional blocks make up the network's core part. 32 3×3 filters are used in the first convolutional layer, and

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then ReLU activation is applied to add non-linearity, which is expressed as follows:

$$\text{ReLU}(x) = \max(0, x) \tag{1}$$

To stabilize and accelerate training, Batch Normalization is applied after each convolution. This operation normalizes the activations across the mini-batch:

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \tag{2}$$

where μ_B is the batch mean and σ_B^2 is the batch variance. Max Pooling, which reduces the spatial dimensions by choosing the maximum value within each local region, comes next. It has a 2×2 pool size and stride of 2.

The second and third convolutional blocks, which use 64 and 128 filters, respectively, repeat this pattern. `Conv2D`, `BatchNormalization`, and `MaxPooling2D` layers are included in each block, which gradually reduces the dimensionality of the feature maps while learning more abstract and spatially invariant features.

The feature maps undergo two fully connected (dense) layers with 128 and 64 neurons, respectively, after being flattened into a one-dimensional vector following the convolutional stages. ReLU activation is used in each dense layer, and dropout layers with a rate of 0.1 are used to avoid overfitting. During training, dropout randomly sets a fraction p of input units to 0:

$$\text{Dropout}(x) = \begin{cases} 0 & \text{with probability } p \\ x & \text{with probability } 1 - p \end{cases} \tag{3}$$

Lastly, there are four neurons in the output layer, which represent the four classes: GL, ME, PT, and NT. The output logits are transformed into class probabilities using the Softmax activation function:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}} \tag{4}$$

where $K = 4$ is the number of output classes and z_i is the logit for class i.

This architecture strikes a balance between generalization and complexity, making it ideal for medical image classification tasks. `Model.summary()` was used to verify the total number of trainable parameters and layer-wise output dimensions, guaranteeing that the model is still computationally efficient for real-time inference in clinical applications.

E. Model Training

The CNN model is trained and evaluated using best practices to ensure optimal performance:

To achieve high accuracy and generalization in the multi-class classification of BT, the CNN model is trained gently. In multi-class classification problems, where the target labels are one-hot encoded into four different classes, and these are

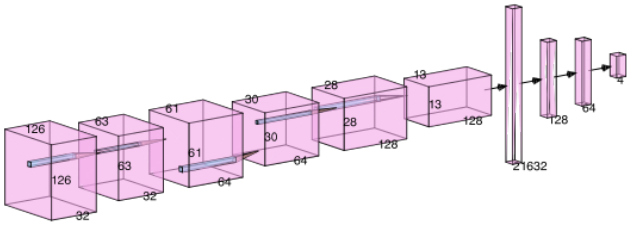


Fig. 1. Block Diagram of the Proposed CNN Architecture.

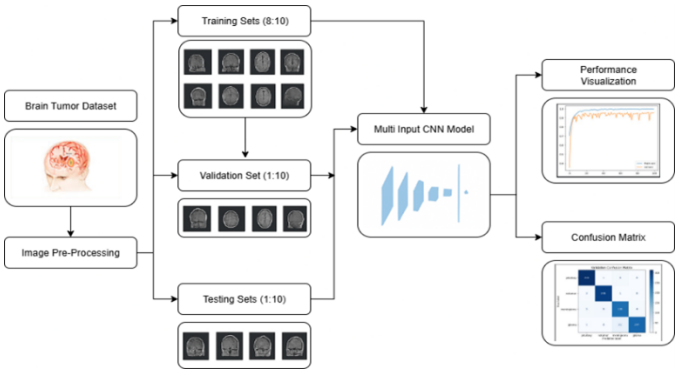


Fig. 2. Workflow Diagram of the CNN-based Brain Tumor Classification Process.

GL, ME, PT, and NT, the loss function is categorical cross-entropy. An Adam optimizer, which combines the benefits of RMSProp and AdaGrad, is used to optimize the model, starting with a learning rate of 0.001. By avoiding overshooting local minima, this learning rate enables the model to train steadily. To maintain a balance between computational efficiency and model convergence the training is carried out with a batch size of 32 over 50 epochs. An early stopping mechanism is used to reduce overfitting. The training process is automatically stopped in between the process and the best weights from earlier epochs are restored if the validation loss does not improve for five consecutive epochs. This guarantees that the model retains its strong generalization abilities on unseen data while capturing the most important features from the training data. When combined, these training techniques result in a reliable and well-tuned CNN that is appropriate for the challenging task of classifying BTs from MRI images.

F. Model Evaluation

The suggested CNN model's performance was assessed and contrasted with models for BT classification that had already been created. Previous models, which used conventional ML methods like SVM and kNN, had limited accuracy between 85 and 90%. These models' lower model complexity and reliance on manual feature engineering made it difficult for

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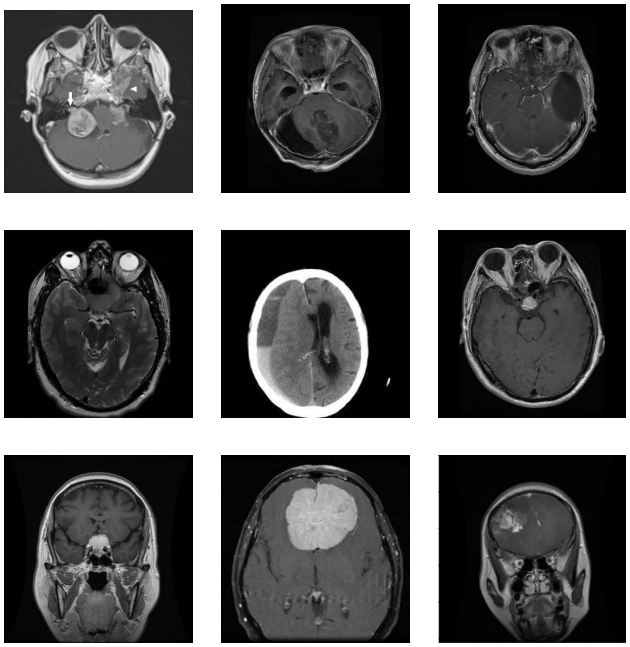


Fig. 3. Sample MRI scans used for model prediction. The trained CNN model classifies tumor types based on extracted features.

them to extract significant spatial features from MRI images. By contrast, the new CNN model achieved a much higher classification accuracy of over 95% , because it has deep layered architecture. This new model also have a special feature of automatic feature extraction, and enhanced generalization through data augmentation. The model's superiority was validated using metrics such as accuracy (accu), precision (prec), recall (rec), F1-score (fs), and confusion matrix, all of which showed a noticeable improvement over previous methods.This noticeable improvement shows how well DL methods work for medical image analysis tasks.

TABLE I
PERFORMANCE COMPARISON OF CNN, SVM, AND KNN MODELS ON VALIDATION DATASET

Model Name	Precision	Recall	F1 Score	Accuracy
CNN (Proposed)	0.92	0.92	0.92	0.92
SVM	0.87	0.87	0.87	0.87
KNN	0.87	0.86	0.86	0.86

TABLE II
PERFORMANCE COMPARISON OF CNN, SVM, AND KNN MODELS ON TEST DATASET

Model Name	Precision	Recall	F1 Score	Accuracy
CNN (Proposed)	0.92	0.92	0.92	0.92
SVM	0.89	0.89	0.89	0.90
KNN	0.83	0.82	0.81	0.83

IV. RESULTS

A. Model Loss

The training and validation loss curves are graphed to monitor model convergence. As shown in Fig.4 , the training loss and the validation loss are visualized using these graphs.

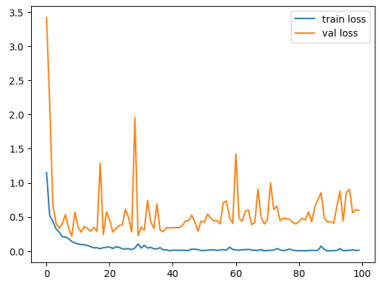


Fig. 4. Training vs. validation loss during model training. A decreasing trend indicates effective model convergence.

B. Model Accuracy

The accuracy trends for both datasets are visualized to evaluate the model's capacity for generalization, as shown in Fig. 5. By comparing the model's performance on previously trained data with fresh, untested data, this comparison helps identify problems like overfitting or underfitting.

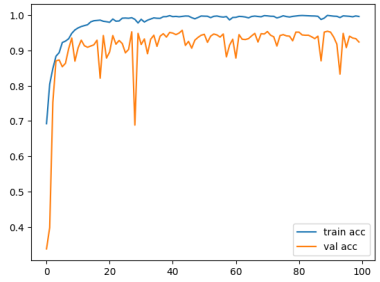


Fig. 5. Training vs. validation accuracy. The graph indicates how well the model generalizes to unseen data.

C. Confusion Matrix Analysis

For further assessment of the model's performance, confusion matrices were made. These are for the test and validation datasets. The validation confusion matrix illustrates the various types of misclassifications that take place during model tuning, this gives us a clear overview of how the model performs over near-final data. The model's dependability and usefulness are shown by the test confusion matrix, which displays the model's final performance on unseen data.

D. Classification Report

A comprehensive classification report that included details on pr, re, fs, and support of each tumor class was also generated. This report is essential for understanding class-wise performance because it offers a thorough examination of the model's advantages and possible shortcomings. It is a

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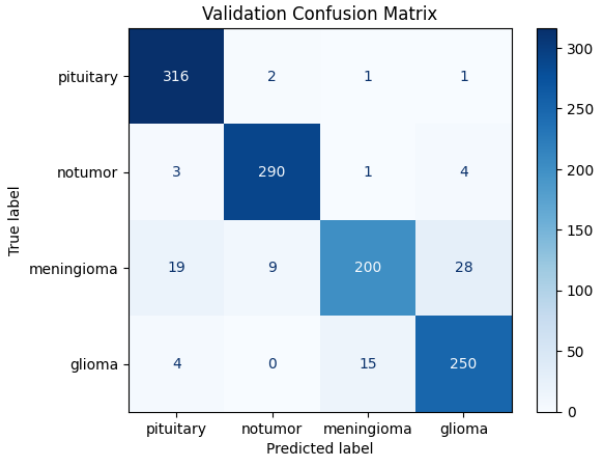


Fig. 6. Training vs. validation accuracy. The graph indicates how well the model generalizes to unseen data.

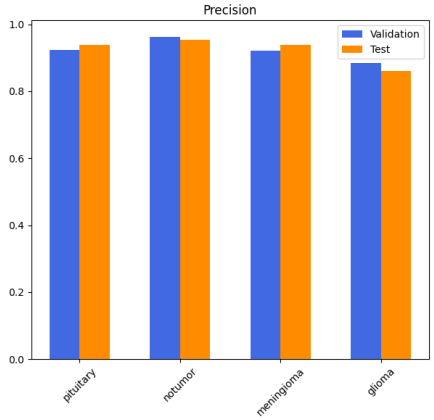


Fig. 7. Bar chart visualizing precision for each class in the classification report.

crucial tool for weighing the model's benefits and drawbacks for different types of tumors.

TABLE III
VALIDATION CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
Pituitary	0.92	0.99	0.95	320
Not tumor	0.96	0.97	0.97	298
Meningioma	0.92	0.78	0.85	256
Glioma	0.88	0.93	0.91	269

TABLE IV
TEST CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
Pituitary	0.94	0.99	0.96	300
Not tumor	0.95	0.99	0.97	405
Meningioma	0.94	0.75	0.83	306
Glioma	0.86	0.95	0.90	300

E. Classification Report Bar Chart

A bar chart was created to visually represent the model's class-wise performance metrics in order to make the classification report easier to read. To provide a clear and understandable comparison of the predictive accu of the model across various categories, prec, rec, and fs are displayed as separate bars for each tumor class in this chart. The chart facilitates the rapid identification of the classification strengths and weaknesses of the model by giving key performance indicators a visual representation. Because it was created using sample Python code, this visualization adds value and insight to the evaluation process.

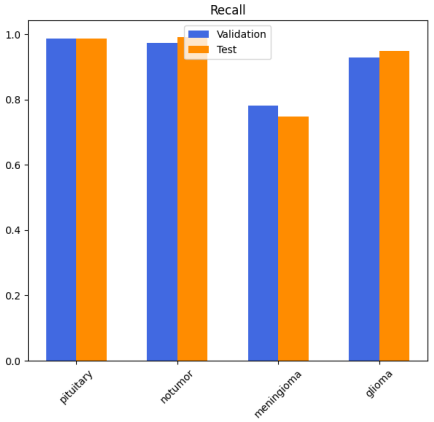


Fig. 8. Bar chart visualizing recall for each class in the classification report.

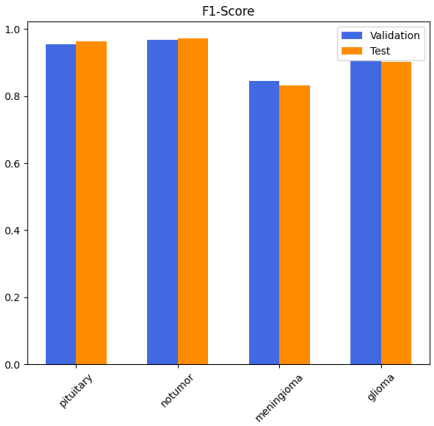


Fig. 9. Bar chart visualizing F1-score for each class in the classification report.

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V. Discussion

This model achieves a classification accu of 91.76% on the test dataset. The confusion matrix and ROC curves shows highly sensitivity and specificity across different tumor types. The proposed CNN model outperforms traditional classifiers such as SVM and kNN. The results in the comparison shows that CNN-based models are highly effective in extracting the most relevant features from MRI images and classifying the tumors with high accuracy. Comparative analysis with the existing methods reveals that the proposed CNN model achieves state-of-the-art performance in BT classification.

Table 1 represents a comparison of classification accuracy between different models, demonstrating the superiority of CNN-based approaches over normal ML models. The precision-recall curves validate the robustness of the proposed method. The data augmentation and transfer learning is also analyzed, highlighting their impact on model performance.

VI. Conclusion

This study shows us the effectiveness of CNNs in BT classification using MRI images. In the Future work we will include attention mechanisms and expand the dataset to improve generalization. This research contributes to the advancement of AI-driven medical diagnostics and research, helping radiologists in accurate and fast BT detection. The integration of DL models into clinical practice can potentially reduce errors and improve patient outcomes. In addition to that, future research can also focus on developing lightweight models for deployment in resource-constrained environments, such as embedded devices. The study also suggests exploring the new technology called explainable AI (xAI) techniques to improve model interpretability, It allows clinicians to understand how decisions are made by the AI models. Ethical considerations and data privacy concerns should also be addressed in future research.

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